whether the universe actually had the highly ordered beginning that this explanation of time's arrow requires.

Cosmology is among the oldest subjects to captivate our species. And it's no wonder. We're storytellers, and what story could be more grand than the story of creation? Over the last few millennia, religious and philosophical traditions worldwide have weighed in with a wealth of versions of how everything—the universe—got started. Science, too, over its long history, has tried its hand at cosmology. But it was Einstein's discovery of general relativity that marked the birth of modern scientific cosmology.

Shortly after Einstein published his theory of general relativity, both he and others applied it to the universe as a whole. Within a few decades, their research led to the tentative framework for what is now called the *big bang theory,* an approach that successfully explained many features of astronomical observations (Chapter 8). In the mid-1960s, evidence in support of big bang cosmology mounted further, as observations revealed a nearly uniform haze of microwave radiation permeating space —invisible to the naked eye but readily measured by microwave detectors—that was predicted by the theory. And certainly by the 1970s, after a decade of closer scrutiny and substantial progress in determining how basic ingredients in the cosmos respond to extreme changes in heat and temperature, the big bang theory secured its place as the leading cosmological theory (Chapter 9).

Its successes notwithstanding, the theory suffered significant shortcomings. It had trouble explaining why space has the overall shape revealed by detailed astronomical observations, and it offered no explanation for why the temperature of the microwave radiation, intently studied ever since its discovery, appears thoroughly uniform across the sky. Moreover, what is of primary concern to the story we're telling, the big bang theory provided no compelling reason why the universe might have been highly ordered near the very beginning, as required by the explanation for time's arrow.

These and other open issues inspired a major breakthrough in the late 1970s and early 1980s, known as *inflationary cosmology* (Chapter 10). Inflationary cosmology modifies the big bang theory by inserting an